

Assignment-2
AI 24304 Machine Learning Techniques Lab
Deadline: 14th August, 2026, 4:00 pm

Question 1:

Use the following dataset to find a simple linear regressor ($y = \beta_0 + \beta_1 x$) with respect to each feature (consider features with only continuous values) in the dataset.

<https://archive.ics.uci.edu/dataset/186/wine+quality>

The dataset has 11 features which can be used to estimate the quality of wine.

- a) Estimate the values of β_0 and β_1 under individual features using gradient descent.
- b) Create a plot of iterative changes in β_0 (x-axis) and β_1 (y-axis) for 100 iterations with z-axis denoting the error convergence (use both mean squared error (MSE) and mean absolute error (MAE) to record the iterative changes). Also record your iterative response in a csv file.
- c) Create a plot of iterative changes in β_0 and β_1 using the stopping criteria $|\beta_{k+1} - \beta_k| \leq 10^{-6}$ (for all the features) with z-axis denoting the error convergence (use both mean squared error (MSE) and mean absolute error (MAE) to record the iterative changes). Also record your iterative response in a csv file.
- d) Estimate coefficient of determination when a feature is used to fit the model.
- e) Check for the normality of residuals. The residuals should be normally distributed. This means that the residuals should follow a bell-shaped curve. If the residuals are not normally distributed, then linear regression will not be an accurate model.
- f) Repeat the steps (a)-(e) for all the features.
- g) Draw your observations.

Note: In-built functions for finding the linear regressor not allowed.

Questions 2:

Repeat Question 1 under multiple linear regression ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \cdots \beta_n x_n$) when all the features are used to estimate the target variable y . Individual plots are not required as all features are used at the same time. Record the gradient updates for all parameters in a csv file.

Note: In-built functions for finding the linear regressor not allowed.

Question 3:

Repeat Question 1 under polynomial linear regression ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 \cdots \beta_n x_1^n$) when all the features are used to estimate the target variable y . Use your domain knowledge/

intuition/discretion to decide the relative importance of individual features. Individual plots are not required as all features are used at the same time. Record the gradient updates for all parameters in a csv file.

Note: In-built functions for finding the linear regressor not allowed.

Instructions:

- a) All the programs must be submitted on google classroom **"ONLY"** within the specified deadline. Kindly stick to the deadline period as no assignment will be accepted beyond the deadline period.
- b) The file names should be same as the question number of the programs. For example, if there are five questions in "Lab Assignment x", name your files as assx-1.py, assx-2.py, assx-3.py, etc; so on and so forth; where x is the assignment number.
- c) All the above files must be wrapped into a zipped folder; with your roll number as its name in a single zipped file (.zip or .rar format). For example, if your roll number is BTECH/15004/24, name your file as "btech1500424-assx.zip" or " btech150042422-assx.rar"; where x is the assignment number.
- d) Always keep a copy of your programs stored somewhere so that it can be presented whenever asked for.
- e) Some questions may ask to note down your observations. You can prepare a PDF document for that purpose and add the same as 'obs-assx.pdf'. You can demarcate your observations for different programming questions within the PDF file.
- f) All the programs will be checked for software similarity with a tolerance limit of 30%.